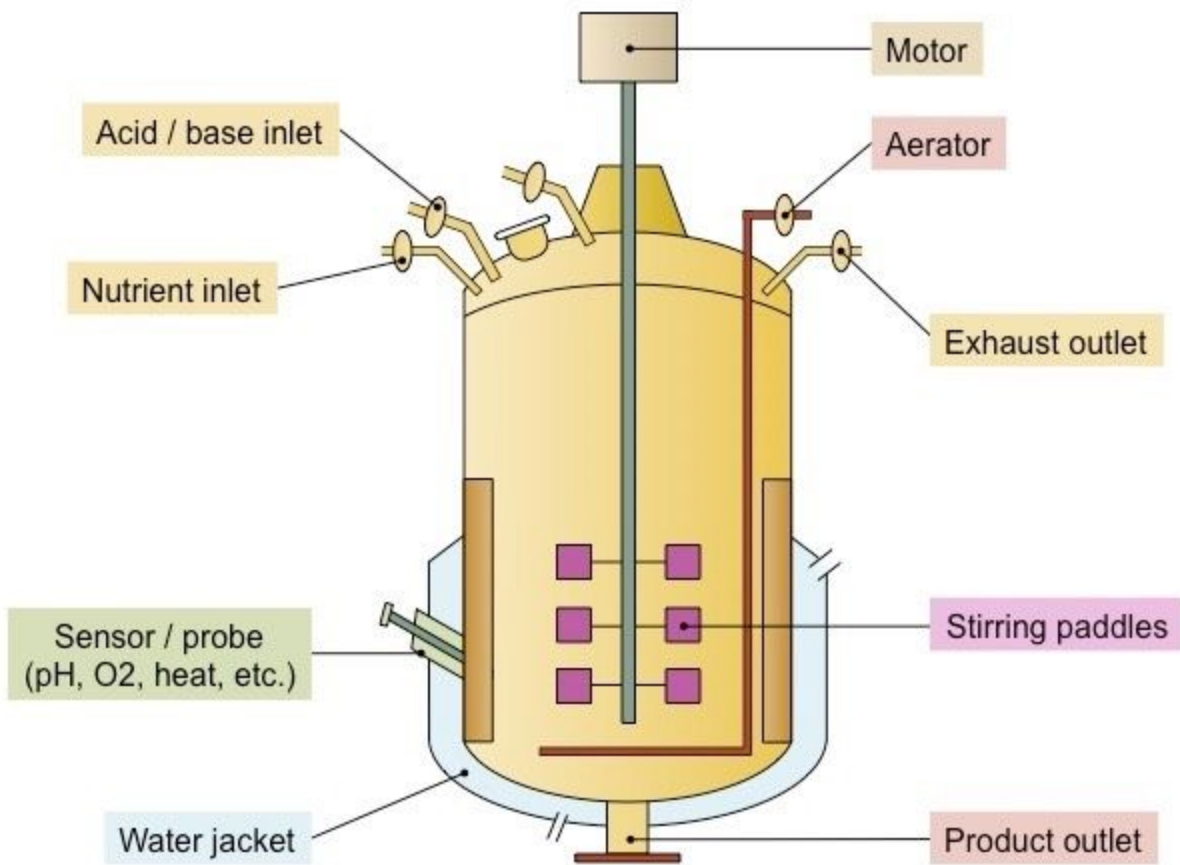


Fermentation Reactor



Names: Eric, Stephon, Rayan, Siva, and Sedra

Reactor Research Project

CHE 321 : Dr. Berry

Introduction

Industrial fermentation is the elegant synthesis of biology and engineering that was present thousands of years ago in ancient civilizations and will continue to revolutionize our world as we go forward. It plays a major role in multi-billion-dollar industries such as food, chemical, and pharmaceutical. The term “fermentation” was initially used to describe the breakdown of molecules in anaerobic conditions to generate energy. Now, “fermentation” is used in industry with a broader scope as the cultivation of microorganisms in a fermenter or bioreactor. The fermenter is a closed vessel that provides controllable and optimal conditions for the microorganisms’ growth. In order to better understand the impact of fermentation to society, it is important to reflect on the history, the current implications technologically and economically, and the future of the fermenter reactor.

Traditional fermentation was used far before the true science of the process was understood. Beer brewing, one of the earliest applications of fermentation, was present in agrarian civilizations 10,000 years ago. Seen as a substitute for water that was often contaminated, beer became a common beverage (Markov 2012). During the French Renaissance, wine making utilized a floating mat of aerobic bacteria while acetic acid (vinegar) was continuously removed from the barrel as a byproduct. In the 19th century, Louis Pasteur demonstrated experimentally that yeast cells convert glucose into ethanol, carbon dioxide, and other molecules anaerobically. Pasteur then developed “pasteurization”, a heat treatment for foods and beverages that destroys pathogenic microorganisms. (Eldrige 2019). During the 20th century, there were several developments in fermenter engineering as a result of the two world wars. During World War I, *Clostridium acetobutylicum* was used to produce acetone from starch. During World War II, *Penecillum* was used to produce penicillin with large-scale fermentation technology

(Meyer 2017). These progressions are seen below in Figure 1. Among many other discoveries, enzymes have become increasingly important in industry as catalysts.

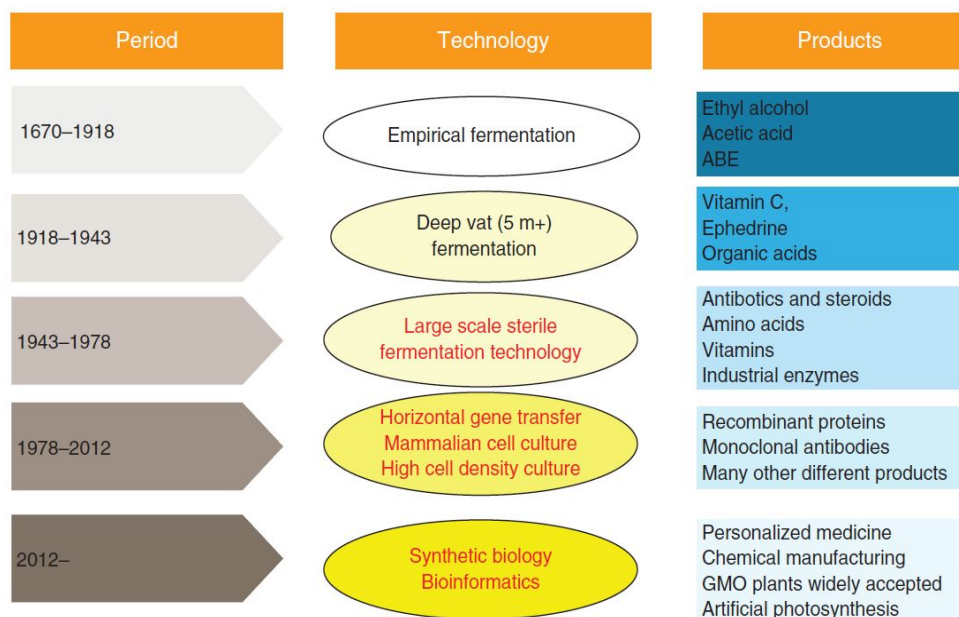


Figure 1: Historical Development of Fermentation Technology

Over 500 companies worldwide utilize fermentation to produce a variety of products. With the introduction of genetic engineering and global issues at the dawn of the 21st century, advances in fermenter technology could revolutionize our world. Genetic traits can now be inserted directly into the microorganism with the use of recombinant DNA (rDNA). Upon finding the adequate microorganism to carry out a specific biological process, it can be modified to increase its efficiency of production and make a process economically competitive (Impacts of Applied Genetics 1981). Even though cutting-edge technology is available, every new problem requires a unique approach to find a solution. Increasing concerns over climate change and scarce resources for a growing global population have caused companies to shift towards more sustainable and economically favorable practices. A vast majority of research is currently being done on biomass production including biofuels. With limited

availability of arable and fertile land, agriculture cannot produce enough biomass to meet the world's energy demands. However, agriculture can provide enough feedstock to the chemical industry. Biomass is used as a source for protein for human food or animal feed, as starter cultures, as a pesticide or fertilizer, and as a fuel source. For example, *Cyanobacterium spirulina* is 50 to 70 percent protein and has been used as a dietary supplement (Markov 2012). This is an ecologically sound and nutrient rich alternative to obtain protein. Livestock requires land, water, feed, and time to mature. Biofuels, such as ethanol and biogas, are considered future substitutes for fossil fuels since they generate less pollutants. Bioethanol is the largest fermentation product in volume and sales (Deloitte 2014). There are currently four regions that will be attractive for fermentation businesses, seen in Figure 2, including:

1. Brazil: sugar cane
2. United States: corn
3. South-East Asia: tapioca/cassava and sugarcane
4. North-West Europe: wheat and sugar beets

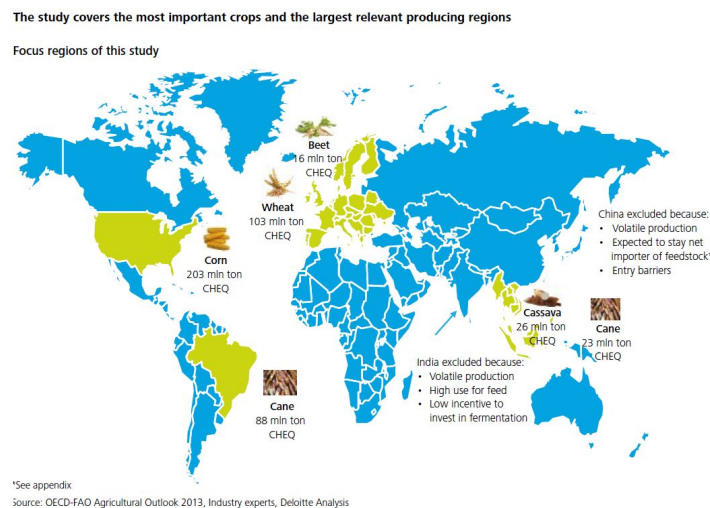


Figure 2: Production Types Across the World

The United States utilizes yeast fermentation of sugar from corn starch to produce ethanol and carbon dioxide. Ethanol is separated from the fermentation broth using distillation. Brazil, the second largest ethanol producer, utilizes sugar cane fermentation and has a more efficient process. Biogas is generated from wastewater and animal waste (organic matter) by bacteria and *Archaea* (Markov 2012). Future investment in industrial fermentation will vary in cost depending on the target product. On average, a non-GMP will cost \$90-\$100 million USD and a GMP will cost \$170-\$180 million USD (Meyer 2017). Future innovation of the fermenter reactor will depend on four drivers:

- finances (economically viable)
- legal regulations
- consumer perception of the product
- new technology development

Detailed Process

Fermentation reactors, known as bioreactors, serve to sustain life for the purpose of creating a biological product. Therefore, a primary focus of bioreactors is to create conditions for life to thrive. These conditions vary based on the organism used in the reactor, which can range from unicellular fungi to bacteria to mammalian cells. In order to keep these organisms living, parameters such as oxygen levels, fragility to shear stress, pH, and heat must be controllable (Jagani 2010). Furthermore, the genetics of the organism itself must be taken into account, as selection factors may alter the course of genetic evolution, especially in rapidly dividing species (Meyer 2017).

Therefore, it is of prime importance to take great care in designing a bioreactor. Over 99% of bioreactors operate with cells suspended in a medium (Meyer 2017). The size of this reactor will depend on the organism and its needs, with ranges from a few mm³ to up to 500 m³ in the industrial setting

(Jagani 2010). A table of volume corresponding to species type is seen below in Figure 3, derived from work by Meyer (2017).

Organism	Average Fermenter Volume (in m ³)
Bacteria	250
Yeast	30-80
Fungi	600
Protozoa	2
Mammalian Cells	25
Stem Cells	<1

Figure 3: Average Bioreactor Volume per Organism Type

After considering size, the mode of operation comes into play. The simplest method is through a batch reactor; however, due to lack of control, fed-batch reactors may be used instead (Meyer 2017). In such reactors, food or other essentials are sent into the reactor to allow for more control over the conditions of the reactor. For enhanced product production, continuous reactors are used, generally CSTRs (Meyer 2017).

In these aforementioned reactors, oxygen availability is an important variable to consider. Oxygen must be transmitted to all the organisms within the mixture without the presence of air bubbles, which may be deadly to sensitive lifeforms (Jagani 2010). Therefore, bubble-free aeration is essential, which has been found to be easily achievable through a CSTR coupled with an oxygen-permeable membrane (Meyer 2017). The constant stirring motion allows for oxygen and heat to spread throughout

the tank, and different devices such as impellers can be used to limit the shear stress on the organisms. As air bubbles hit the bottom of the discs, the impellers displace the oxygen throughout the mixture (Jagani 2010). One popular such device in aerobic fermentation is the Rushton turbine, as it can quickly displace oxygen without air bubbles through axial propulsion of fluid (Meyer 2017). However, this limits the vertical mixing of the fluid, which can be corrected for by axial flow impellers (Jagani 2010).

This also leads to the importance of ensuring the proper height of the reactor; if the reactor is too long, it may lead to limited oxygen availability at the bottom, as well as large hydrostatic pressure (Meyer 2017). With this in mind, the accepted diameter to height ratio is 1:3 when bacteria is the organism, and 2:3 for cell cultures. (Meyer 2017) This difference is due to the greater dependence on heat and oxygen mixing for microbial organisms, compared to greater vulnerability to shear stress in mammalian cells (Meyer 2017). Another limiting variable is foaming, which can damage filters. Thus, in many systems, mechanical foam breakers may be used (Jagani 2010).

While the above considerations allow for a deeper look into the processing details of bioreactors, another factor that must be considered is their incorporation into larger plants. A decision must be made to create plants with a single product or multiproduct. Single product plants focus largely on the process, due to the necessity to create a standardized approach to creating the desired product across the plant (Meyer 2017). In contrast, multiproduct plants must be flexible to allow for quick changes to reactor design to allow for multiple products to be made. Furthermore, contamination is a large concern in multiproduct plants, due to the presence of multiple organisms and substrates that may inhibit other processes (Meyer 2017). Overall, single product plants will be more efficient, however, multiproduct plants do offer more versatility.

Although efficiency is a prime factor as to why, many industries have switched to fermentation reactors for their chemical processes, the economic value is another strong drive. A sundry of industries utilize fermentation, such as companies that create household appliances, fragrances, and pharmaceuticals. Industrial-scale fermentation generates over \$300,000,000,000 per year. In the United States, one meter cubed of fermentation produces over \$95,000 every day (Meyer 2017). The most prominent industries that have fermentation reactors as an integral part of their process is the biotechnological industry. This industry can be divided into two sectors, known as *red* and *white* as red biotech works with antibodies, parenteral, and therapeutic proteins, while *white* biotech is an umbrella term for other applications. Biofuels, generic chemicals, industrial enzymes, and vitamins are examples of other products that depend heavily on fermentation reactors. However, bioethanol is one of the largest products produced by fermentation by volume.

These massive industries all choose fermentation over other reactors for several reasons. From a financial viewpoint, fermentation is a great investment. As it is shown above the profit per meter cubed is massive. Manufacturing prices are low and much cheaper than extraction and chemical synthesis. Fermentation reactors can be implemented in a company in two ways, either at a large scale or in a lab setting. Large scale reactors need to follow Good Manufacturing Practice (GMP) while small scale lab settings can lead operation with non-GMP standards. GMP is an industry-wide system that upholds high-quality standards. Pharmaceuticals must follow GMP to ensure patient safety. Abiding by GMP guidelines can seem very expensive, but to many, it is considered an investment that has large benefits. GMP requires plants to have special equipment, clean rooms, and heating, ventilation, and air conditioning (HVAC) units. Non-GMP sites can be much cheaper to start up, but the extra precautions are taken to ensure sterile work environments.

Beyond the economic draw, fermentation leads to more sustainable sourcing and manufacturing. This leads to another aspect that makes fermentation appealing. It is deemed as an environmentally friendly technique. Pharmaceutical businesses are often under fire for the amount of waste active pharmaceutical ingredients (APIs) generate. As the world grows and the demand for antibiotics, drugs, and vaccines grows, so do the repercussions of production such as waste. In these times and with these demands, fermentation provides a sustainable solution to this dilemma. Many industries are adopting fermentation reactors for the perception that they can be environmentally friendly.

Nonetheless, many companies choose fermentation for its ability to prosper innovation. Fermentation reactors can be used as models for the next steps in gene transfer. Many biotech companies use fermentation reactors for synthetic biology and bioinformatics, which can lead to artificial photosynthesis and personalized medicine, which is the future of how diseases will be treated (Betts 2006).

Once companies consider the multitude of benefits that can be gained from using fermentation reactors, the next step is to understand what kind of fermentation reactor would be ideal and how it would be operated. The main goal of this method is that they need to be safe and reproducible. Additionally, they must be done in a sterile and clean environment as many of these products come into direct contact with humans. When scaling up fermentation processes, the four key aspects that plants analyze are pressure, pH, temperature, and oxygen levels. This, the controls and monitoring systems for these characteristics are required.

The two most prominent options are either single-use technology or stainless steel. Single-use technology does not require any cleaning, rarely has contamination, and is much cheaper, however, only a limited quantity of products can be generated, and it is a relatively new technology that many

industries do not know how to handle. The other option would be stainless steel fermentation reactors, yet this also has its ups and downs. Stainless steel reactors can handle large capacities, have been used in plants for decades, and can be controlled easily, but they are costly, require extensive cleaning, and can not be moved easily (Clapp 2016).

Each design has its benefits and drawbacks, but ultimately when companies choose to use fermentation reactors, regardless of their design, a thorough cost analysis needs to be conducted as shown below in Figure 4.

Initial Costs	Material Energy Personnel Maintenance and repair (external) External services and general expenses
Continuous Cost	Utilities
Overhead Cost	Administration Quality Control / Quality Assurance Sales and Marketing R&D

Figure 4: Analyzing Cost during Scale-up (Meyer 2017)

Design Equations

Fermentation reactors are special types of **Catalytic Reactors** known as **Bioreactors**. For these types of reactors, instead of a typical catalyst found in most other reactors, there is an enzyme. Enzymes are the acting catalyst for Bioreactor processes. For fermentation, the reactors can be run via several processes.

Chemostat (Continuous) Fermenter

1. Biomass Concentration Equations / Fermentation (Conversion) Time

The Chemostat is a type of **Continuous Process** in which the growth of a culture is limited by one specific nutrient cell. Starting with the mass balance, we derive the following relation:

$$In - Out + Gen. - Cons. = Acc. \rightarrow Gen - Consumption = Accumulation$$

Where Inlet and Outlet are FX_0 and FX , respectively, **Generation** = $\mu_g * X * V_R$, **Consumption** = $k_d * X * V_R$, and Accumulation = $V_R \frac{dX}{dt}$. Plug into the mass balance equation to reach:

$$FX_0 - FX + V_R \mu_g X - V_R k_d X = V_R \left(\frac{dX}{dt} \right)$$

Where the variables are:

- F = Nutrient Solution Flow Rate
- V_R = Volume of Culture in Tank
- X_0 & X = Initial and Final Biomass Cell Concentration, respectively.
- μ_g = Cell Growth Rate
- k_d = Cell Death Rate/Endogenous Metabolism

The material balance can be further simplified into:

$$\frac{dX}{dt} = DX_0 + (\mu_g - k_d - D)X$$

and D = Dilution Rate, which equals: $D = \frac{F}{V_R}$. Furthermore, we can assume $X_0 = 0$ since in most cases, the initial Feed is completely sterilized: $\frac{dX}{dt} = (\mu_g - k_d - D)X$. To simplify, note that $(\mu_g - k_d) = \mu_{net}$ which is the **Net Specific Growth Rate**.

Unsteady State

Separate to reach: $\frac{dX}{X} = (\mu_{net} - D) dt$ and integrate further to arrive at:

$$\ln \frac{X}{X_0} = (\mu_{net} - D)t$$

If we want to solve for the final Cell Biomass Concentration, we arrive at:

$$= X_0 e^{(\mu_{net} - D)t}$$

Steady State

If we operate the simplified mass balance under steady state, we see that:

$$(\mu_{net} - D)X = 0 \text{ (although } X \neq 0, \text{ so } \mu_{net} = D).$$

2. Substrate Utilization Equations

If a growth rate limiting substrate is present, the material balance around the substrate is shown to be:

$$V_R \left(\frac{dS}{dt} \right) = FS_0 - FS - \frac{V_R \mu_g X}{Y_{X/S}^M} - \frac{V_R q_p X}{Y_{P/S}} - m_s X V_R$$

Where the following variables are:

- S_0 and S = Initial and Final Substrate Concentration, respectively.
- $Y_{X/S}^M$ = Maximum Yield Coefficient of Cell Biomass Growth (X) $\equiv \frac{\text{grams of } X}{\text{grams of Substrate, } S}$
- q_p = Rate of Product Formation $\equiv \frac{\text{grams of } P}{\text{grams of } X \cdot \text{unit time}}$
- $Y_{P/S}$ = Yield Coefficient of Product Formation (P) $\equiv \frac{\text{grams of } P}{\text{grams of Substrate, } S}$

- $m_s = \text{Substrate Maintenance Coefficient} \equiv \frac{\text{grams of } S}{\text{grams of } X \cdot \text{unit time}} = \frac{k_d}{Y_{X/S}^M}$

The above can be simplified by noting that $\frac{\mu_g}{Y_{X/S}^M} - \frac{q_p}{Y_{P/S}} - m_s = q_s$ where q_s is the **Rate of Substrate**

Consumption. Divide both sides by V_R , which simplifies the Substrate equation into:

$$\left(\frac{dS}{dt}\right) = D(S_0 - S) - q_s X$$

If we assume Steady State and no product formation, the above equation becomes:

$$\frac{D+k_d}{Y_{X/S}^M} = \frac{D(S_0-S)}{X} \rightarrow \frac{D}{Y_{X/S}^M} + m_s = \frac{D}{Y_{X/S}^{AP}}$$

Where $\frac{1}{Y_{X/S}^{AP}} = \text{Apparent Yield Coefficient of Cell Biomass Growth} = \frac{S_0-S}{X}$

Note that we re-expanded q_s to show the above terms.

3. Formation of Product in a Chemostat

Starting with the mass balance: $In - Out + Gen. - Cons. = Acc. \rightarrow In - Out + Gen = Accumulation$,

where **In** = 0, **Out** = **FP**, **Gen.** = $q_p * X * V_R$, and **Acc.** = $V_R \frac{dP}{dt}$,

We arrive at the following balance:

$$-FP + q_p X V_R = V_R \left(\frac{dP}{dt}\right)$$

Divide by V_R :

$$-D(P) + q_p X = \left(\frac{dP}{dt}\right)$$

Assuming Steady State: $D(P) = q_p X \rightarrow P = \frac{q_p X}{D}$

Note that, under ideal conditions, a Chemostat is nearly identical to a CSTR.

Batch Fermentation

1. **Biomass Concentration Equations / Fermentation (Conversion) Time**

The reactor performance for a Batch Fermenter is largely dependent on how the **fermentation/conversion time** is. Starting with the mass balance in a Batch Fermenter, we see that:

$$In - Out + Gen. - Cons. = Acc. \rightarrow Gen - Consumption = Accumulation$$

Inlet and Outlet are 0 since this is a batch process, **Generation** = $\mu_g * X * V_R$, **Consumption** = $k_d * X * V_R$, and Accumulation = $V_R \frac{dX}{dt}$. Plug into the mass balance equation to reach:

$$\mu_g X V_R - k_d X V_R = V_R \frac{dX}{dt}$$

Divide by V_R and separate:

$$(\mu_g - k_d) dt = \frac{dX}{X}$$

Integrate:

$$(\mu_g - k_d) t = \ln \left| \frac{X}{X_0} \right|$$

After solving for X we get:

$$X = X_0 e^{(\mu_g - k_d)t}$$

Remember again that $\mu_g - k_d = \mu_{net}$

$$X = X_0 e^{\mu_{net} t}$$

We can derive the **Fermentation (Conversion) Time** by manipulating the above to reach:

$$t = \frac{1}{\mu_g} * \ln \ln \frac{X}{X_0}$$

2. **Substrate Utilization Equations**

Start with the same mass balance equation:

$In - Out + Gen. - Cons. = Acc. \rightarrow Consumption = Accumulation$ since here we don't have any Input,

Output, or Substrate Generation. **Consumption** = $q_s * X * V_R$ and **Accumulation** = $V_R \frac{dS}{dt}$. Plugging this into the mass balance, we see:

$$\left(\frac{\mu_g}{Y_{S}^M} - \frac{q_p}{Y_{S}^P} - m_s \right) X V_R = V_R \frac{dS}{dt}$$

Divide by V_R and separate:

$$\left[\left(\frac{\mu_g}{Y_{S}^M} - \frac{q_p}{Y_{S}^P} - m_s \right) X_0 e^{\mu_{net} t} \right] dt = dS$$

Integrate:

$$\left[\left(\frac{\mu_g}{Y_{S}^M} - \frac{q_p}{Y_{S}^P} - m_s \right) \frac{X_0}{\mu_{net}} \right] e^{(\mu_{net})t} - 1 = S - S_0$$

Where here, μ_m is the **Maximum Specific Growth Rate**. Using the above equation, the Final Substrate Concentration can be equated as:

$$S_0 + \left[\left(\frac{\mu_g}{Y_{S}^M} - \frac{q_p}{Y_{S}^P} - m_s \right) \frac{X_0}{\mu_{net}} \right] e^{(\mu_{net})t} - 1 = S$$

Under these conditions **Fermentation (Conversion) Time** is:

$$t = \frac{1}{\mu_{net}} \ln \left[1 + \frac{S - S_0}{\left[\left(\frac{\mu_g}{Y_{S}^M} - \frac{q_p}{Y_{S}^P} - m_s \right) \frac{X_0}{\mu_{net}} \right]} \right]$$

3. Formation of Product in a Batch Reactor

Starting with the mass balance in a Batch Fermenter, we see that:

$$\begin{aligned} In - Out \text{ Gen.} - Cons. &= Acc. \rightarrow Gen = Accumulation . \text{ **Generation** } = q_p * X * V_R \text{ and **Accumulation** } \\ &= V_R \frac{dP}{dt} \end{aligned}$$

where:

$$q_p X V_R = V_R \frac{dP}{dt}$$

Divide by V_R and separate:

$$q_p X_0 e^{\mu_{net} t} dt = dP$$

Integrate and the Final Product Concentration can be equated as:

$$\frac{q_p X_0 e^{\mu_{net} t}}{\mu_{net}} - 1 = P$$

Under these conditions **Fermentation (Conversion) Time** is:

$$t = \frac{1}{\mu_{net}} \ln \left[[P + 1] \frac{\mu_{net}}{X_0 q_p} \right]$$

The batch material balance can be modified to suit conditions as needed, such as Steady State, no Biomass Death Rate ($k_d = 0$, which in turn would make $\mu_{net} = \mu_g$)

Fed Batch Fermentation

Fed-Batch (aka Semi-Continuous) Fermentation is a type of fermentation which can be considered as a combination between **Continuous** and **Batch** fermentation. There is a continuous/intermittent feeding of nutrient into the fermentation tank, but **Volume is not constant** ($F = \frac{dV_R}{dt} \rightarrow V_R = V_{R0} + Ft$) unlike **Batch** and **Continuous Reactors**. It operates at **quasi-steady state**.

4. Biomass Concentration Equations / Fermentation (Conversion) Time

Starting with the mass balance, we derive the following relation:

$$In - Out + Gen. - Cons. = Acc. \rightarrow In + Gen = Accumulation$$

Where **Inlet** = $F X_0$, **Generation** = $\mu_g * X * V_R$, **Consumption** = $k_d * X * V_R$, and **Accumulation** = $\frac{d(XV_R)}{dt}$. Plug into the mass balance equation to reach:

$$F X_0 + V_R \mu_g X - V_R k_d X = X \frac{dV_R}{dt} + V_R \frac{dX}{dt}$$

Rearranging the appropriate values and dividing by V_R :

$$D(X_0 - X) + \mu_{net} X = \frac{dX}{dt}$$

Remember that we are at **Quasi-Steady State** and we assume sterile feed ($X_0 = 0$):

$$X(\mu_{net} - D) = 0 \rightarrow X \neq 0, \text{ so } \mu_{net} = D$$

Which we can see is identical to the expression obtained when **Continuous Fermentation** is at Steady State.

There is an additional parameter. While the Biomass **Concentration** is constant, the **Mass** of the Biomass Cells does change depending on the Volume. If we take $X = x V_R$ we see that

$$\frac{dX}{dt} = \frac{d(xV_R)}{dt} \rightarrow V_R \frac{dx}{dt} + x \frac{dV_R}{dt}$$

Take $\frac{dx}{dt} = 0$, remember that $F = \frac{dV_R}{dt}$, and $x = SY_{\frac{x}{s}}$, from Yield Equation :

$$\frac{dX}{dt} = x \rightarrow \frac{dX}{dt} = SY_{\frac{x}{s}}^{AP} F$$

Integrate:

$$X_{max} = X_0 + \left(SY_{\frac{x}{s}}^{AP} F \right) t$$

5. Substrate Utilization Equations

The mass balance for Fed-Batch is almost identical to that of Continuous Fermentation as shown below. Note, there is no outlet flow here and remember Volume is not constant:

$$V_R \left(\frac{dS}{dt} \right) + S \frac{dV_R}{dt} = F S_0 - q_s X V_R$$

We are under Quasi-Steady State. Divide both sides by V_R , and input $F = \frac{dV_R}{dt}$ which simplifies the Substrate equation into:

$$D(S_0 - S) - q_s X = 0 \rightarrow D(S_0 - S) = q_s X$$

6. Formation of Product in Fed-Batch Reactor

Starting with the mass balance: $In - Out + Gen. - Cons. = Acc. \rightarrow Gen = Accumulation$, where

Gen. = $q_p * X * V_R$, and **Acc.** = $V_R \frac{dP}{dt} + P \frac{dV_R}{dt}$, We arrive at the following balance:

$$q_p X V_R = V_R \frac{dP}{dt} + P \frac{dV_R}{dt}$$

Divide by V_R and substitute the correct values:

$$q_p X = F P + \left(\frac{dP}{dt} \right)$$

Remember we are at Quasi-Steady State: $\frac{q_p X}{F} = P$

Example Solutions

Chemostat Continuous Fermenter Reactor Example Solution:

A culture of bacteria that secretes a product within a chemostat continuous fermenter reactor is operated at steady state. The specific growth rate of the mass is described using the Monod equation, and the rate at which the product forms is described using the Leudeking-Piret kinetics. The feed to the reactor is pure and the flow rates at the inlet and exit can be assumed to be constant. The following information is known for this system:

$$\mu_{max} = 0.5 \text{ hours}^{-1}$$

$$K_S = 0.4 \frac{\text{g}}{\text{L}}$$

$$Y_{X/S} = 0.5$$

$$Y_{P/S} = 0.7$$

$$\alpha = 0.4$$

$$\beta = 0.1 \text{ hours}^{-1}$$

$$S_o = 7 \text{ g/L}$$

$$F = 11 \text{ L/hours}$$

$$V = 300\text{L}$$

$$\text{Monod Equation: } \mu = \frac{\mu_{max}S}{K_S+S}$$

$$\text{Leudeking-Piret kinetics: } r_P = (\alpha\mu + \beta)X$$

A. Derive the steady state mass balances for S, X, and P.

B. Solve for the steady state concentration of S.

C. Solve for the steady state concentration of X.

D. What is the productivity rate of the substrate per unit time and what is it a function of?

A. Begin by writing out mass balance for each term, then replace with given rates.

Mass balance on S:

$$In - Out + Gen. - Cons. = Acc.$$

$$F(S_o - S) + r_s V = 0$$

$$F(S_o - S) - V \left[\frac{1}{Y_{X/S}} \cdot \frac{\mu_{max} S}{K_S + S} \cdot X + \frac{\left(\alpha \frac{\mu_{max} S}{K_S + S} + \beta \right) X}{Y_{P/S}} \right] = 0$$

Mass balance on X:

$$In - Out + Gen. - Cons. = Acc.$$

$$-FX + \mu XV = 0$$

$$-FX + \frac{\mu_{max} S}{K_S + S} XV = 0$$

Mass balance on P:

$$In - Out + Gen. - Cons. = Acc.$$

$$-FP + r_p V = 0$$

$$-FP + \left(\alpha \frac{\mu_{max} S}{K_S + S} + \beta \right) XV = 0$$

B. From the mass balance on X from part A.

$$-FX + \frac{\mu_{max} S}{K_S + S} XV = 0$$

[Rearrange and solve for S].

$$-FX + \frac{\mu_{max} S}{K_S + S} XV = 0$$

$$FX = \frac{\mu_{max} S}{K_S + S} XV$$

$$F = \frac{\mu_{max} S}{K_S + S} V$$

$$FK_S + FS = \mu_{max} SV$$

$$\frac{FK_S + FS}{V} = \mu_{max} S$$

$$\frac{FK_S}{V} = S \left(\mu_{max} - \frac{F}{V} \right)$$

$$S = \frac{\frac{FK_S}{V}}{(\mu_{max} - \frac{F}{V})}$$

$$S = \frac{\frac{11 \frac{L}{hours} - 0.4 \frac{g}{L}}{300L}}{\left(\frac{0.5}{hours} - \frac{11 \frac{L}{hours}}{300L} \right)}$$

$$S = 3.17 \times 10^{-2} \frac{g}{L}$$

C. From the mass balance on S from part A.

$$F(S_o - S) - V \left[\frac{1}{Y_{X/S}} \cdot \frac{\mu_{max} S}{K_S + S} \cdot X + \frac{\left(\alpha \frac{\mu_{max} S}{K_S + S} + \beta \right) X}{Y_{P/S}} \right] = 0$$

$$F(S_o - S) = V \left[\frac{1}{Y_{X/S}} \cdot \frac{\mu_{max} S}{K_S + S} \cdot X + \frac{\left(\alpha \frac{\mu_{max} S}{K_S + S} + \beta \right) X}{Y_{P/S}} \right]$$

$$\frac{F(S_o - S)}{V} = \left[\frac{1}{Y_{X/S}} \cdot \frac{\mu_{max} S}{K_S + S} \cdot X + \frac{\left(\alpha \frac{\mu_{max} S}{K_S + S} + \beta \right) X}{Y_{P/S}} \right]$$

$$\frac{F(S_o - S)}{V} = \left[\frac{1}{Y_{X/S}} \cdot \frac{\mu_{max} S}{K_S + S} + \frac{\left(\alpha \frac{\mu_{max} S}{K_S + S} + \beta \right)}{Y_{P/S}} \right] \cdot X$$

$$X = \frac{\frac{F(S_o - S)}{V}}{\left[\frac{1}{Y_{X/S}} \cdot \frac{\mu_{max} S}{K_S + S} + \frac{\left(\alpha \frac{\mu_{max} S}{K_S + S} + \beta \right)}{Y_{P/S}} \right]}$$

$$X = \frac{\frac{11}{\text{hours}} \left(7 \frac{\text{g}}{\text{L}} - 3.17 \times 10^{-2} \frac{\text{g}}{\text{L}} \right)}{300L} + \frac{\left(\frac{0.5}{\text{hours}} \cdot 3.17 \times 10^{-2} \frac{\text{g}}{\text{L}} + \frac{0.1}{\text{hours}} \right)}{0.4 \frac{\text{g}}{\text{L}} + 3.17 \times 10^{-2} \frac{\text{g}}{\text{L}}}$$

$$X = 0.285 \frac{\text{g}}{\text{L}}$$

- D. The organism will grow at the rate equal to the dilution rate, the productivity rate is a function of how quickly the organism can grow.

$$Productivity = Y_{P/S} \cdot \mu = Y_{P/S} \cdot \frac{\mu_{max} S}{K_S + S}$$

$$Productivity = 0.7 \frac{\text{grams of product}}{\text{grams of substrate}} \cdot \frac{\frac{0.5}{\text{hours}} 3.17 \times 10^{-2} \frac{\text{g}}{\text{L}}}{0.4 \frac{\text{g}}{\text{L}} + 3.17 \times 10^{-2} \frac{\text{g}}{\text{L}}}$$

$$Productivity = 0.0257 \frac{\text{grams of product}}{\text{grams of substrate} \cdot \text{hours}}$$

Batch Fermentation Reactor Example Solution:

Determine how long it takes for a biomass concentration to decrease by 90% within a batch fermentation reactor. The net generation rate is provided below along with the reaction info.

$$\mu_{net\ generation\ rate} = \frac{\mu_{generation\ rate} \cdot S}{K_S + S} - k_{death\ rate}$$

$$\mu_{generation\ rate} = 0.25\ hours^{-1}$$

$$k_{death\ rate} = 0.3\ hours^{-1}$$

$$K_S = 0.1\ g/L$$

$$S = 4\ g/L$$

Begin by writing the mass balance for the biomass (X):

$$In - Out + Gen. - Cons. = Acc.$$

$$\mu_g X V_R - k_d X V_R = V_R \frac{dX}{dt}$$

$$\mu_g X - k_d X = \frac{dX}{dt}$$

$$X(\mu_g - k_d) = \frac{dX}{dt}$$

$$\mu_{net} = \mu_g - k_d = \frac{\mu_{generation\ rate} \cdot S}{K_S + S} - k_{death\ rate}$$

$$X \left(\frac{\mu_{\text{generation rate}} \cdot S}{K_S + S} - k_{\text{death rate}} \right) = \frac{dX}{dt}$$

$$\left(\frac{\mu_{\text{generation rate}} \cdot S}{K_S + S} - k_{\text{death rate}} \right) = \frac{1}{X} \frac{dX}{dt}$$

[Integrate both sides.]

$$\left(\frac{\mu_{\text{generation rate}} \cdot S}{K_S + S} - k_{\text{death rate}} \right) t = \ln \left(\frac{X}{X_0} \right)$$

$$\left(\frac{\mu_{\text{generation rate}} \cdot S}{K_S + S} - k_{\text{death rate}} \right) t = \ln \left(\frac{X_0(1-x)}{X_0} \right)$$

$$t = \frac{\ln \left(\frac{X_0(1-x)}{X_0} \right)}{\left(\frac{\mu_{\text{generation rate}} \cdot S}{K_S + S} - k_{\text{death rate}} \right)}$$

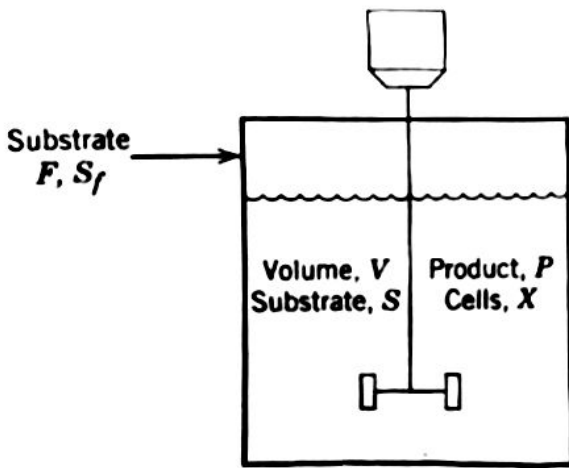
$$t = \frac{\ln(1-x)}{\left(\frac{\mu_{\text{generation rate}} \cdot S}{K_S + S} - k_{\text{death rate}} \right)}$$

$$t = \frac{\ln(1-0.90)}{\left(\frac{0.25 \frac{1}{\text{hours}} \cdot 4 \frac{\text{g}}{\text{L}}}{0.1 \frac{\text{g}}{\text{L}} + 4 \frac{\text{g}}{\text{L}}} - 0.3 \frac{1}{\text{hours}} \right)}$$

$$t = \frac{\ln(1-0.90)}{\left(\frac{0.25 \frac{1}{\text{hours}} \cdot 4 \frac{\text{g}}{\text{L}}}{0.1 \frac{\text{g}}{\text{L}} + 4 \frac{\text{g}}{\text{L}}} - 0.3 \frac{1}{\text{hours}} \right)}$$

$$t = 41.0 \text{ hours}$$

Fed Batch Fermentation Example Solution



Consider a Fed-Batch fermenter reactor that is operating at a quasi steady state. The feed is sterile and the initial concentration of biomass is 0.4 g/L. Given the information below, determine the cell death rate.

$$\mu_{net} = \frac{\mu_{max} \cdot S}{K_S + S} - k_{death\ rate}$$

$$\mu_{max} = 0.01\ hours^{-1}$$

$$K_S = 0.1\ g/L$$

$$S = .3\ g/L$$

$$F = 2\ L/hours$$

$$V_R = 500L$$

Begin by performing a mass balance on the biomass.

$$In - Out + Gen. - Cons. = Acc.$$

$$FX_0 + V_R\mu_g X - V_R k_d X = \frac{d(XV_R)}{dt}$$

$$FX_0 + V_R\mu_g X - V_R k_d X = X \frac{dV_R}{dt} + V_R \frac{dX}{dt}$$

$$\mu_{net} = \mu_g - k_d$$

$$FX_0 + V_R\mu_{net} X = X \frac{dV_R}{dt} + V_R \frac{dX}{dt}$$

$$F = \frac{dV_R}{dt}$$

$$FX_0 + V_R \mu_{net} X = FX + V_R \frac{dX}{dt}$$

$$F(X_0 - X) + V_R \mu_{net} X = \frac{dX}{dt}$$

$$\frac{F}{V_R}(X_0 - X) + \mu_{net} X = \frac{dX}{dt}$$

$$D = \frac{F}{V_R}$$

$$D(X_0 - X) + \mu_{net} X = \frac{dX}{dt}$$

[Using the quasi-steady state assumption.]

$$D(X_0 - X) + \mu_{net} X = 0$$

[Feed is sterile.]

$$X_0 = 0$$

$$D(X_0 - X) + \mu_{net}X = 0$$

$$D(X_0 - X) + \mu_{net}X = 0$$

$$\mu_{net} = D$$

$$\frac{\mu_{max} \cdot S}{K_S + S} - k_{death\ rate} = D$$

$$k_{death\ rate} = \frac{\mu_{max} \cdot S}{K_S + S} - D$$

$$k_{death\ rate} = \frac{\mu_{max} \cdot S}{K_S + S} - \frac{F}{V_R}$$

$$k_{death\ rate} = \frac{0.01 \frac{L}{hours} \cdot 0.3 \frac{g}{L}}{0.1 \frac{g}{L} + 0.3 \frac{g}{L}} - \frac{2 \frac{L}{hours}}{500L}$$

$$k_{death\ rate} = 0.0185\ hours^{-1}$$

Safety Considerations

Safety is essential for the plant operators, the microorganisms in the fermenter reactor, and the consumer.

With respect to the plant, it must follow:

- biosafety requirements (normally BSL1 and BSL2)
- safety requirements for highly potent active pharmaceutical ingredients (HPAPI)
- current good manufacturing practice (cGMP) regulations enforced by the FDA

With respect to the microorganisms, it must account for:

- oxygen demand
- sensitivity to shear
- sensitivity to process and culture variations
- sensitivity to variations within the bioreactor

These are general guidelines but may vary based on the type of microorganisms and will be discussed in detail.

The plant must adhere to several guidelines to maintain safe operating conditions, especially for the plant operators. The biosafety levels (BSLs) for industrial fermentation plants are normally BSL1 and BSL2 (BSL3 and BSL4 are also possible). A thorough risk assessment of the plant is used to determine the appropriate biosafety level and should be reviewed and revised routinely. BSL1 has low risk microbes and risk increases to BSL4 which has high risk microbes, seen in Figure 5 below. A BSL

will determine the standard operating procedures (SOPs), the personal protective equipment (PPE), and the plant equipment necessary to have a safe work environment. Training is also required for all employees to ensure safe laboratory practices and procedures (Laboratory Safety Manual 2004). HPAPI and cGMP are important in the pharmaceutical industry since the products are usually high value and purity is the top priority. This leads to stricter guidelines for the operating conditions and equipment which also determines the capital investment, time for approval, and running cost. These regulations are enforced to protect the plant operator, consumer, and environment.

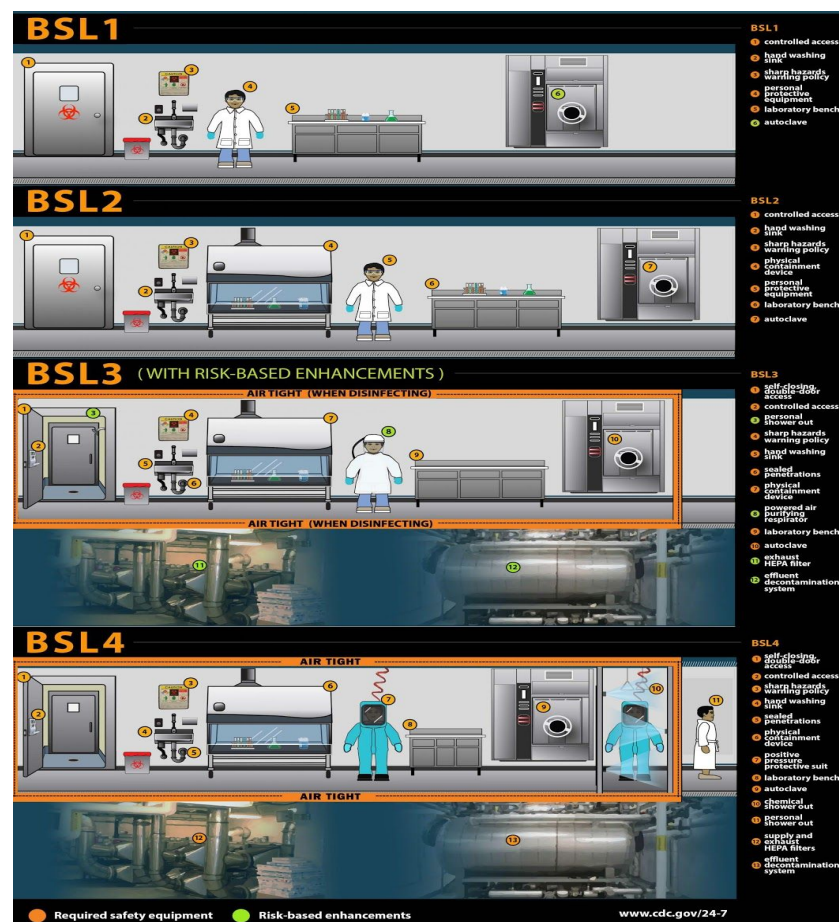


Figure 5: Biosafety Levels of Industrial Fermentation Plants

The plant must also ensure that the fermenter design is suitable for the microorganisms. Due to the sensitive nature of the microorganisms, the large-scale process requires control of several parameters that were previously discussed so that the process is safe, consistently efficient, easily reproducible, reliable, and cost-effective (Meyer 2017). These parameters (temperature, pressure, dissolved oxygen, pH) are regulated with online controls. For plants that produce high value products, critical process parameters (CPPs) and critical quality attributes (CQAs) can aid in reducing the risk of batch failure. The heating/cooling system can include a jacket or external coils. Pressure gauges and safety valves prevent excess pressure. The pH and dissolved oxygen are measured with probes (Pumphrey 1996). The main safety concern when working with fermenter reactors is the risk of contamination. In order to sterilize a fermenter, heated steam is maintained at a high temperature (usually 120-122 °C) for at least 30 minutes. This is a critical step since temperature can destroy the medium or lead to loss of nutrient quality. Filter sanitation is a good alternative for very heat sensitive components. It is later cooled with water injected into the jacket. Compressed, filter sterilized process air is supplied through a sparger (Jagani 2010). The exhaust air leaves through a heat exchanger that prevents backward contamination. All feeds are added as sterile solutions. The tank should be fully drainable to recover product and be easily cleaned. It may be necessary to perform a cleaning in place (CIP) with treated water and a sterilization in place (SIP) with clean steam routinely. Horizontal pipes are not allowed since material can accumulate. The number of openings should be limited, and the seals should fit properly to avoid leakage. An impeller or anti foaming agent should be used to prevent foaming (Meyer 2017).

Further Information:

Market overview for key fermentation products in 2013 and annual growth projection until 2020

Category	Market Size in product output (quantity produced)	Average theoretical yield	Market size in carbohydrate input required	Market size in value	Average added value generated from carbohydrate	Market growth until 2020	Arable land use*
	Mln ton	Ton product/ton glucose	Mln ton CHEQ	Bn USD	USD/CHEQ	% CAGR	Mln ha
Alcohols	99.8	0.51	195.1	110.0	164	4.4%	25.08
Amino Acids	7.1	0.92	7.8	11.0	1,010	5.6%	1.00
Organic Acids	2.9	1.05	2.8	3.5	850	8.8%	0.36
Biogas	0.1	0.27	0.5	0.2	0	5.0%	0.06
Polymers	0.2	0.93	0.2	0.6	2,600	13.5%	0.03
Vitamins	0.2	0.96	0.2	0.7	3,100	2.6%	0.03
Antibiotics	0.2	1.00	0.2	0.8	3,600	4.0%	0.03
Industrial Enzymes	0.1	1.00	0.1	0.3	2,600	8.0%	0.01
Total	110.5		206.8	127.0		4.6%	26.6

*Land use estimates vary greatly with the different region and the crop used to produce the sugar, current estimates are based on weighted average yields (ton CHEQ / ha) of sugar cane and sugar beet

Note: Rough-cut added value from carbohydrate obtained by market value – average cost of CHEQ at 400 USD per ton and ignoring side-streams, energy, etc.

Source: BCC Research, FO Licht Renewable Chemicals Database, NOVA Institut, FAO/OECD, Deloitte Analysis

Global Arable Land
1,500 mln ha

Determining the market size product output, product value, and market growth are major factors for future fermenter investment. Currently, alcohol dominates in volume and value. Polymer technology is projected to have the greatest growth in the coming years.

Currently eight key categories of molecules or compounds are produced commercially and ample opportunities for future developments exist

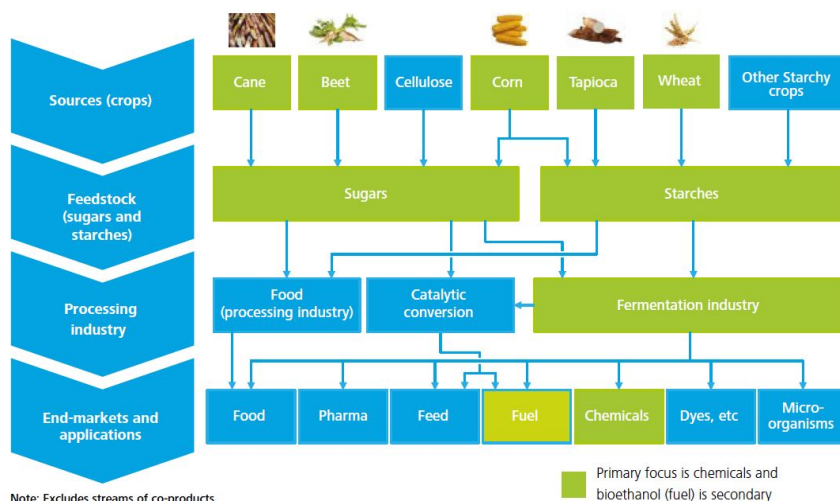
Currently commercial fermentation processes							
Alcohols & Ketones	• Ethanol • Butanol • BDO • Acetone	Organic acids	• Citric • Lactic • Succinic	Polymers	• Xanthan • PHA	Anti-biotics	• Beta-lactam • Tetracycline • Clavulic acid
Amino acids	• MSG • Lysine • Threonine • Tryptophan	Biogas	• Methane	Vitamins	• Vitamin C • Vitamin B2 • Vitamin B12	Industrial enzymes	• Amylase • Cellulase • Lipase • Protease

Selection of future developments based on current research (ranging from theoretical research to testing plant phase)*							
Alkanes	• Nonane • Tetra-decane	Olefins	• Butadiene • Isoprene • Propene • Farnesene • Biodiesel	Amines	• Histamine • Tyramine	Esters	• Malonyl - ACP
Dyes	• Various dyes (e.g. Indigo)	Microbial oils					

*Selection based on interviews and in-depth research that showed these products are currently produced on a very small scale, e.g. farnesene is produced on laboratory scale of appr. 0.04 million tons per year
 Note: Yeasts and other microorganisms excluded from the scope of this study
 Source: BCC Research, FO Licht Renewable Chemicals Database, Deloitte Analysis

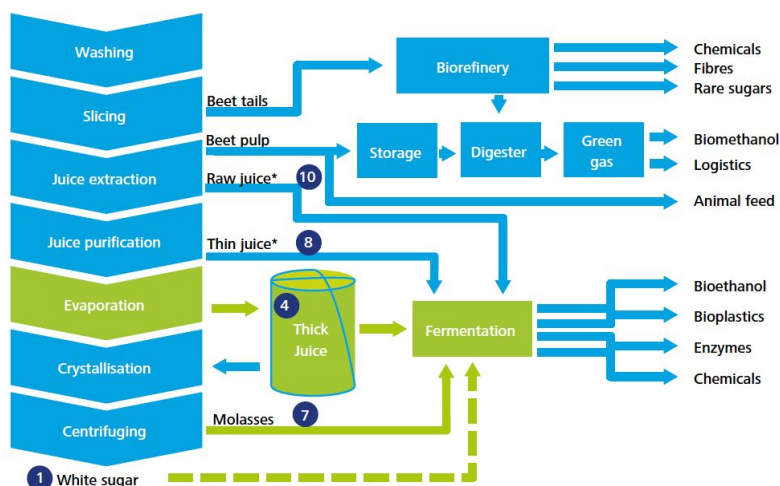
The table shows categories and specific examples. A lot of research is being done on biodiesel since it is expected to become a substitute for fossil fuels in energy production.

Chemicals derived from cane, beet, corn, tapioca, and wheat through fermentation are the primary focus of this study



Major crops from different parts of the world are used as feedstock for industrial fermentation. The flow diagram shows the general overview from crop to final market where the product is used.

The production of sugar from sugar beets has several side streams that can be used to produce amongst others bioethanol and chemicals



* Raw juice and thin juice for fermentation not possible outside beet campaign due to decay
Note: Technological advances could result in new routes to fermentation
Source: Suiker Unie

A specific example for beets shows how they are processed. Each intermediate product can serve a different purpose demonstrating the crop's versatility in industry.

Theoretical yields of fermentation products

Category	Compound	Reaction	Molar mass	Theoretical yield**
			(g/mol)	(g product/ g glucose)
Alcohols	Ethanol	$C_6H_{12}O_6 \rightarrow 2 C_2H_5OH + 2 CO_2$	46.07	0.51
	Beer	$C_6H_{12}O_6 \rightarrow 2 C_2H_5OH + 2 CO_2$	46.07	0.51
	Wine	$C_6H_{12}O_6 \rightarrow 2 C_2H_5OH + 2 CO_2$	46.07	0.51
	Isobutanol	$C_6H_{12}O_6 \rightarrow C_4H_9OH + 2 CO_2 + H_2O$	74.12	0.41
	PDO	$2 C_6H_{12}O_6 \rightarrow 3 C_7H_8O_2 + 3 CO_2$	76.09	0.63
	N-butanol	$C_6H_{12}O_6 \rightarrow C_4H_9OH + 2 CO_2 + H_2O$	74.12	0.41
	BDO	$11 C_6H_{12}O_6 \rightarrow 12 C_8H_{16}O_2 + 18 CO_2 + 6 H_2O$	90.12	0.55
	Acetone	$2 C_6H_{12}O_6 \rightarrow 3 C_3H_6O + 3 CO_2 + 3 H_2O$	58.08	0.48
	Fatty alcohols	Estimated based on typical alcohol yield	Variable	0.50
	Lysine	$7 C_6H_{12}O_6 + 12 NH_3 \rightarrow 6 C_6H_{12}N_2O_2 + 6 CO_2 + 18 H_2O$	146.19	0.70
Amino Acid	MSG*	$5 C_6H_{12}O_6 + 3 O_2 + 6 NH_3 \rightarrow 6 C_5H_9NO_2 + 12 H_2O$	169.11	1.13
	Glutamic acid	$5 C_6H_{12}O_6 + 3 O_2 + 6 NH_3 \rightarrow 6 C_5H_9NO_2 + 12 H_2O$	147.13	0.98
	Threonine	$2 C_6H_{12}O_6 + 3 NH_3 \rightarrow 3 C_4H_9NO_2 + 3 H_2O$	119.12	0.99
	Tryptophan	$23 C_6H_{12}O_6 + 24 NH_3 \rightarrow 12 C_{11}H_{17}N_2O_2 + 6 CO_2 + 102 H_2O$	204.23	0.59
	Citric acid	$2 C_6H_{12}O_6 + 3 O_2 \rightarrow 2 C_6H_8O_7 + 4 H_2O$	192.12	1.07
Organic Acid	Lactic acid	$C_6H_{12}O_6 \rightarrow 2 C_3H_6O_3$	90.08	1.00
	Gluconic acid	$C_6H_{12}O_6 + O_2 \rightarrow C_6H_{12}O_7 + H_2O$	196.16	1.09
	Itaconic acid	$5 C_6H_{12}O_6 + 3 O_2 \rightarrow 6 C_4H_4O_5 + 12 H_2O$	130.10	0.87
	Succinic acid	$7 C_6H_{12}O_6 + 6 CO_2 \rightarrow 12 C_4H_6O_4 + 6 H_2O$	118.09	1.12

*MSG is produced by reacting glutamic acid with NaOH, the stoichiometry from sugar is thus the same as for glutamic acid

**Based on reaction stoichiometry, molecular mass of product, and molecular mass of glucose (180.16 g/mol), e.g. for ethanol: 2 ethanol of 46.07 g/mol divided by 1 glucose of 180.16 g/mol equals a yield of 0.51

Source: Deloitte Analysis

Category	Compound	Reaction	Molar mass	Theoretical yield**
			(g / mol)	(g product / g glucose)
Biogas	Methane	$C_6H_{12}O_6 \rightarrow 3 CH_4 + 3 CO_2$	16.04	0.27
Polymers	Xanthan	$70 C_6H_{12}O_6 + 27 O_2 \rightarrow 12 C_{35}H_{49}O_{29} + 126 H_2O$	933.75	0.89
	PHA*		Variable	1.00
Vitamins	Vitamin C	$C_6H_{12}O_6 + O_2 \rightarrow C_6H_8O_6 + 2 H_2O$	176.12	0.98
	Isoascorbic Acid	$C_6H_{12}O_6 + O_2 \rightarrow C_6H_8O_6 + 2 H_2O$	176.12	0.98
	Vitamin B2	$17 C_6H_{12}O_6 + 6 O_2 + 24 NH_3 \rightarrow 6 C_{17}H_{20}N_4O_6 + 78 H_2O$	376.36	0.74
	Vitamin B12	$275 C_6H_{12}O_6 + 336 NH_3 + 24 H_3PO_4 + 24 Co \rightarrow$ $24 C_{63}H_{88}CoN_{14}O_{14}P + 138 CO_2 + 1134 H_2O$	1,355.37	0.66
	Antibiotics*		Variable	1.00
Enzymes	Enzymes*		Variable	1.00
Future developments	Isobutene	$C_6H_{12}O_6 \rightarrow C_4H_8 + 2 CO_2 + 2 H_2O$	56.11	0.31
	Butadiene	$11 C_6H_{12}O_6 \rightarrow 12 C_4H_6 + 18 CO_2 + 30 H_2O$	54.09	0.33
	Isoprene	$7 C_6H_{12}O_6 \rightarrow 6 C_5H_8 + 12 CO_2 + 18 H_2O$	68.12	0.32

* Variable molecular structure and formula, therefore not a single reaction can be defined for these compounds. Since these markets are relatively small, the yields have been set at 1.00 ton product / ton glucose for comparison

**Based on reaction stoichiometry, molecular mass of product and molecular mass of glucose (180.16 g/mol)

Source: Deloitte Analysis

General reactions for products ranging from alcohols, that dominate the market, to future developments.

Sources

“Batch Reactor Operation || Biomass production || Substrate consumption || Product formation ||.” *Youtube*, uploaded by Smart Learning Academy, 20 Jan 2018, <https://www.youtube.com/watch?v=iiWITtZrNnI>

Clarke, Kim G., “Chapter 7: Microbial kinetics during batch, continuous, and fed-batch processes.” *Bioprocess Engineering: An Introductory Engineering and Life Science Approach*. Woodhead Publishing, 2013.

“Continuous Reactor Operation || Biomass production || Substrate consumption || Product form.” *Youtube*, uploaded by Smart Learning Academy, 22 Jan 2018, <https://www.youtube.com/watch?v=iiWITtZrNnI>

Eldridge, Alison. “Louis Pasteur's Contributions to Science.” *Encyclopædia Britannica*, Encyclopædia Britannica, Inc., www.britannica.com/story/louis-pasteurs-contributions-to-science.

“Fed Batch Reactor Operation || Biomass production || Substrate consumption || Product formation ||.” *Youtube*, uploaded by Smart Learning Academy, 21 Jan 2018, <https://www.youtube.com/watch?v=iiWITtZrNnI>

“Genetic Engineering and the Fermentation Technologies.” *Impacts of Applied Genetics: Micro-Organisms, Plants, and Animals*, by J. E. Legates, Congress of the United States, Office of Technology Assessment, 1981, pp. 49–56.

“Infographic: Biosafety Lab Levels.” *Centers for Disease Control and Prevention*, Centers for Disease Control and Prevention, 25 Oct. 2016, www.cdc.gov/cpr/infographics/biosafety.htm.

-
- Jagani , Hitesh. “An Overview of Fermenter and the Design Considerations to Enhance Its Productivity .” *Pharmacologyonline* , Manipal University, 2010,
pharmacologyonline.silae.it/files/newsletter/2010/vol1/27.Iagati.pdf.
- Laboratory Biosafety Manual* . World Health Organization , 2004,
www.who.int/csr/resources/publications/biosafety/Biosafety7.pdf.
- Markov, Sergei. “Industrial Fermentation.” *ResearchGate*, Austin Peay State University, May 2012,
www.researchgate.net/publication/281716235_Industrial_fermentation.
- Meyer, Hans-Peter, et al. “Industrial-Scale Fermentation.” *Industrial Biotechnology Products and Processes*, First ed., Wiley-VCH Verlag GmbH & Co. KGaA, 2017, pp. 3–53,
application.wiley-vch.de/books/sample/3527341811_c01.pdf.
- “Opportunities for the Fermentation-Based Chemical Industry.” *Deloitte*, Sept. 2014,
www2.deloitte.com/content/dam/Deloitte/nl/Documents/manufacturing/deloitte-nl-manufacturing-opportunities-for-the-fermentation-based-chemical-industry-2014.pdf.
- Pumphrey, Brian, and Christian Julien. “An Introduction to Fermentation.” *Wakenbtech*, May 1996,
www.wakenbtech.co.jp/wp/wp-content/uploads/2015/11/nbs_fermentation_basics.pdf.
- Shuler, Michael L., et al. “Chapter 6: How Cells Grow.” *Bioprocess Engineering: Basic Concepts*, 3rd ed., Prentice Hall, 2017.